

Multiple Choice (10 marks)

1. The half-life of radon is 3.80 days. After how many days will only one-sixteenth of a radon sample remain?
 - (a) 26.6 days
 - (b) 19.0 days
 - (c) 3.80 days
 - (d) 5.7 days
 - (e) 22.8 days
2. What is the intensity of the electric field 10 cm from a negative point charge of 500 stat-coulombs in air?
 - (a) 5 dyne/stat-coul
 - (b) 8 dyne/stat-coul
 - (c) 0.5 dyne/stat-coul
 - (d) 100 dyne/stat-coul
 - (e) 10 dyne/stat-coul
3. Calculate the mean ionic activity of a 0.0350 mNa₃PO₄ solution for which the mean activity coefficient is 0.685 .
 - (a) 0.0612
 - (b) 0.0426
 - (c) 0.0753
 - (d) 0.0905
 - (e) 0.0294
4. determine the ratio of the radius of a uranium-238 nucleus to the radius of a helium-4 nucleus.
 - (a) 4.8
 - (b) 7.1
 - (c) 5.5
 - (d) 1.8
 - (e) 4.2
5. The Pleiades is an open star cluster that plays a role in many ancient stories and is well-known for containing ... bright stars.

- (a) 5
- (b) 7
- (c) 9
- (d) 12

True / False (10 marks)

Answer True or False

6. True or False: The correct answer to 'What equal positive charges would have to be placed on Earth and on the Moon to neutralize their gravitational attraction?' is ' 4.210^{13}C '.
7. True or False: The presence of precious metals does not necessarily indicate that a rock is a meteorite.
8. True or False: The potential energy of a compressed spring and a charged object depends on the work done on them during their respective processes.
9. True or False: A completely submerged object displaces a pressure equal to the weight of the fluid above it, affecting its buoyancy.
10. True or False: A voltage of 6 volts will produce a current of 2 amperes in a circuit with a resistance of 3 ohms, according to Ohm's law.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Calculate the net electric flux through a Gaussian cube with an edge length of 55 cm, centered around a particle with a charge of 1.8 C. Discuss the principles involved in your solution.
12. A block of mass 2 kg slides with an initial speed of 4 m/s and compresses a spring with a spring constant of 6 N/m. Analyze the energy transfer and determine the maximum compression of the spring.

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